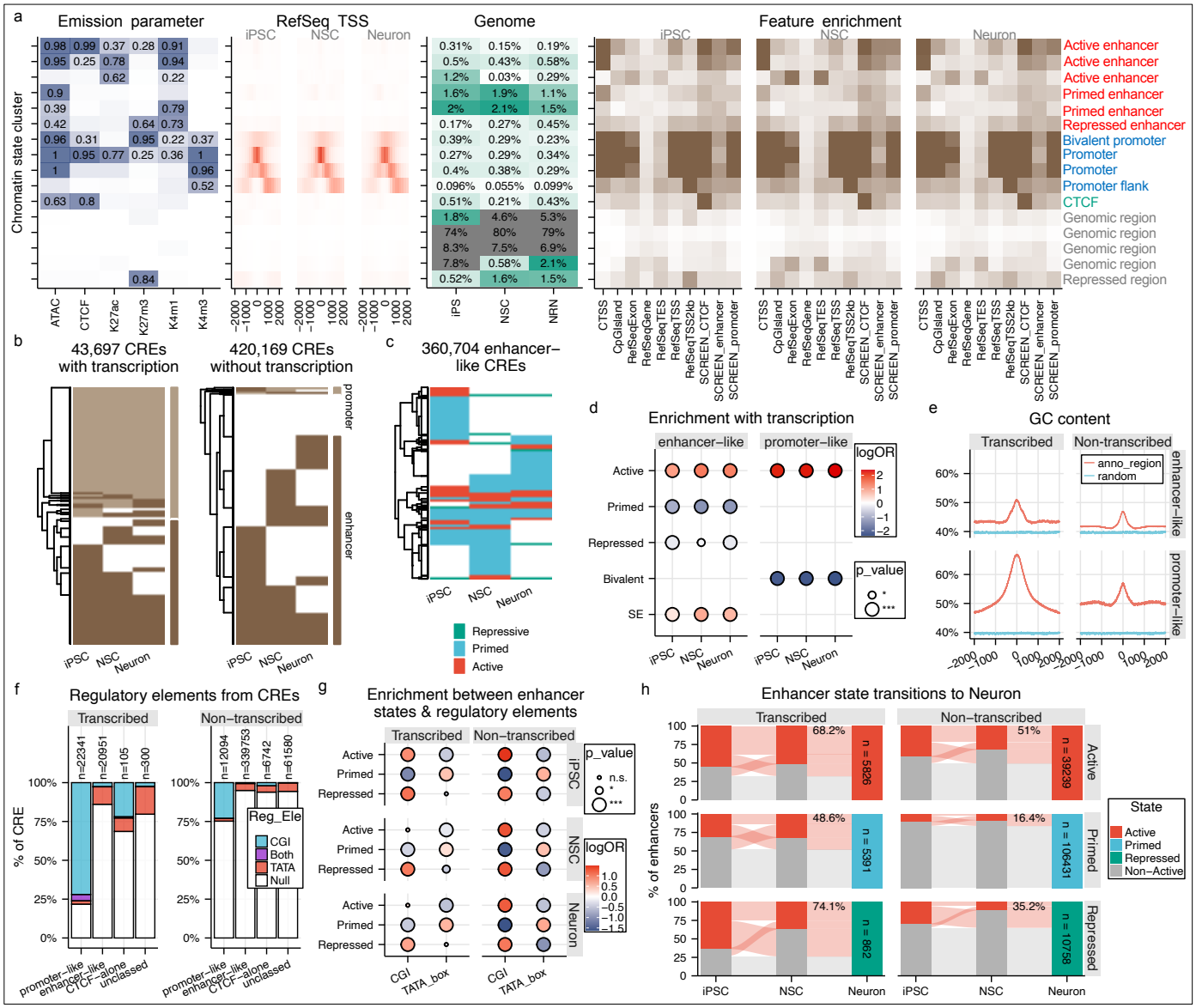


Figure 2 | Chromatin states of CREs correlate with transcription output across differentiation.

a, Genome-wide chromatin state clustering of iPSC, NSC and Neuron determined by ChromHMM using scATAC-seq (grouped by sample) and CUT&Tag data on H3K4m1, H3K4m3, H3K27ac, H3K27m3 and CTCF. Chromatin states were manually curated based on emission profiles. **b**, Heatmaps illustrating the promoter-typing of the CREs, grouped by the presence or absence of TSS signal across the three samples. **c**, Heatmap of all enhancer-like CREs categorized by chromatin states as active, repressive, primed and others (CTCF or genomic regions) across differentiation. **d**, Enrichment between cell-type specific transcription states and chromatin states of enhancer-like and promoter-like CREs, with only accessible CREs in each cell-type included. **e**, GC content of CREs grouped as transcribed and non-transcribed, and promoter-like and enhancer-like. The x-axis represents the distance (nt) from the CRE mid-point. **f**, Percentage of CREs containing specific regulatory elements, including CpG islands (CGI) and TATA boxes. **g**, Enrichment between regulatory elements (CGI and TATA box) and cell-type specific chromatin states of enhancer-like CREs. Tests were performed in sub-groups of transcription states and cell-types. **h**, Number and percentage of neural enhancer-like CREs with active chromatin state in at least one of the three samples across differentiation. Enhancer states (Active, Primed and Repressed) from Neuron are presented separately. Fisher's exact test was used in (**d**) and (**g**) to calculate odds ratios (OR). $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.